



Seven Deadly Sins

Final Presentation

ENGR 7A: Fall 2024

Team Members and Roles

Roles	Names
Team Captain	Derek La
CAD Solidworks Designers	Thu Nguyen & Ashley Vincent
Electronics	Richard Nguyen
Fabrication and Assembly	Yuki Matsuoka
Research and Calculations	Danny Bao
Media & Video Editor	Maggie Dueñas

OBJECTIVES

GOALS FOR ROVER:

- Functional
- Durable
- Lightweight
- Safe



Project Requirements

Safety

- Gear guard
- Wires + connectors insulated away
- Sanded any sharp protruding parts

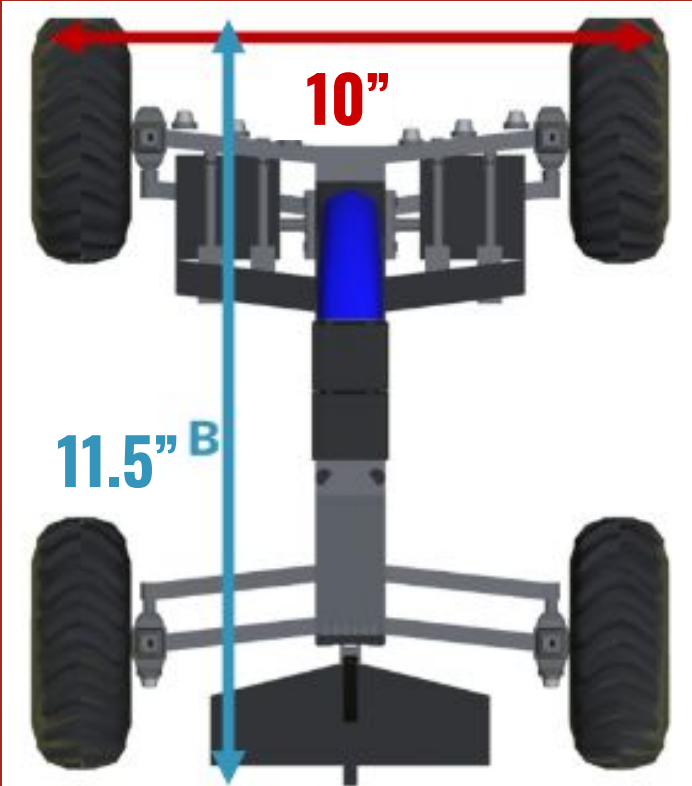


Electronics

- 7.2V battery
- Battery easily removable
- Toggle switch is accessible



Size



Structure

- Self-designed chassis and steering system
- Have TA's approval for designs

Cost

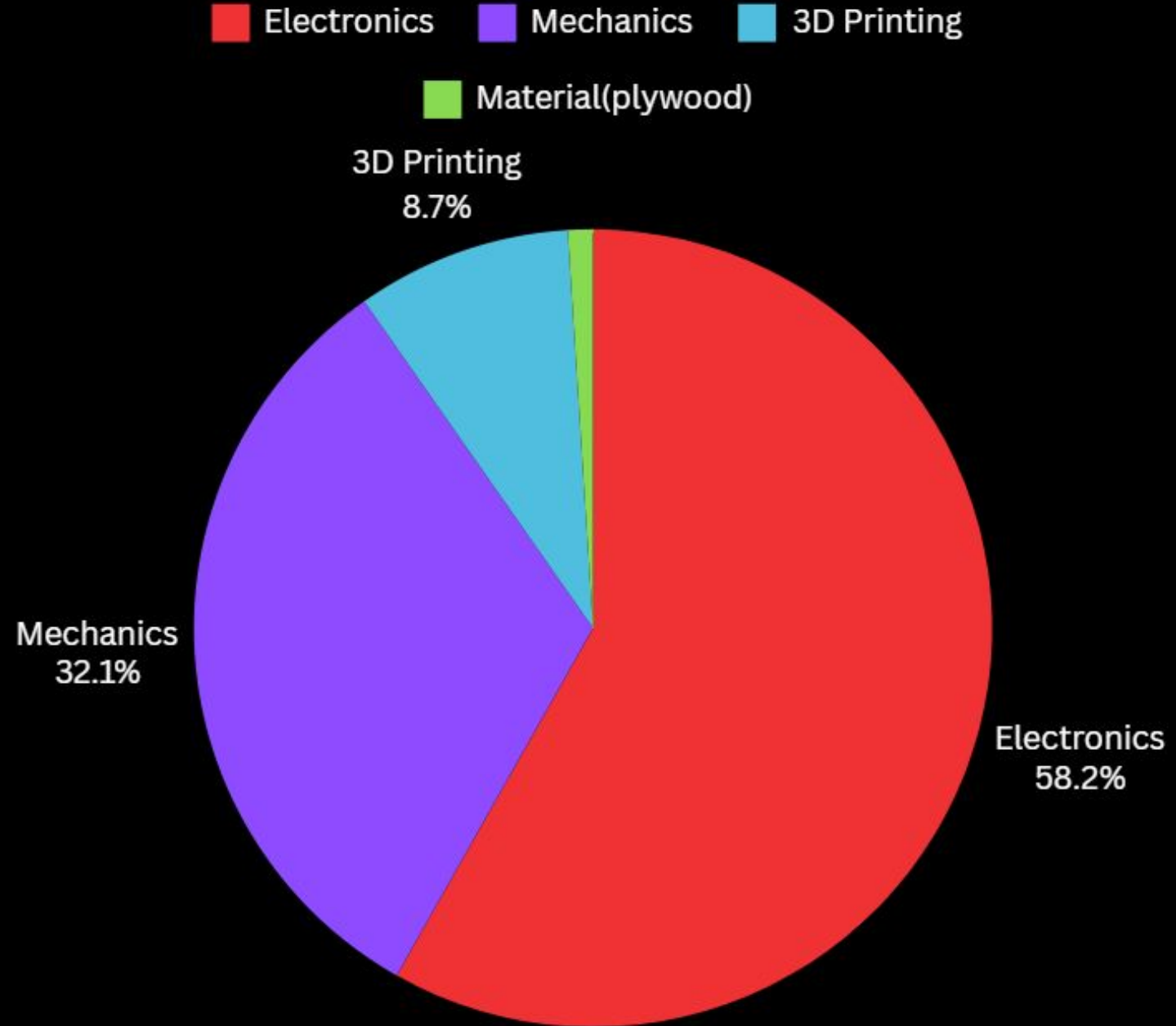
- Complied with \$300 max budget

TOTAL ORDER PRICE =
\$296.89 (including tax)



Structural Expenses:

- 1 Plywood Sheet = \$2.78(1%)
- 6 hrs of 3D Printing (\$24 for steering mechanism)



Final Parts List

- **Electronics**

- 4.4:1 Gearmotor
- Servo
- Brushed ESC
- Buck Converter
- Transmitter + Receiver
- Battery

- **Hardware**

- Motor Bracket
- Servo Horn (25T)
- D-Shaft (6mm bore, 50mm length)
- D-Shaft (6mm bore, 200mm length)
- Sleeve Bearing (6mm bore)
- Shaft Collar (6mm bore)
- Wheel Adapter (6mm bore)
- Gear Hub (6mm bore)
- Pinion Gear (4mm bore, 15T)
- 1/4" birch baltic plywood 12"x12"

- **Miscellaneous**

- Velcro Straps
- Zip Ties
- Various M3 Screws, Nuts, Standoffs
- Glue (wood and gorilla)

Wheel: 85mm

Gear: 52T

Timeline & Major Milestones (WEEK 1-7)

Week (1-4):

- Team Roles
- Steering Research
- Chassis Design
- PO Form



Week (5-7):

- Finalized CAD:
 - Parts
 - Assembly
 - Drawings
- Begin Fabrication
- Preliminary presentation

Calculations for
Ackerman Steering

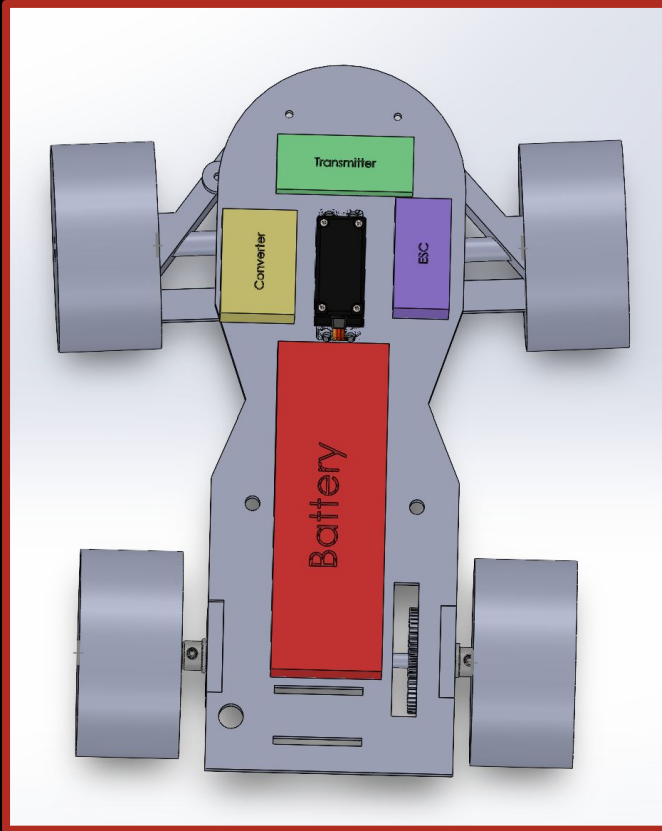
- Solidworks
CAD issues
- Member illness

Challenges:

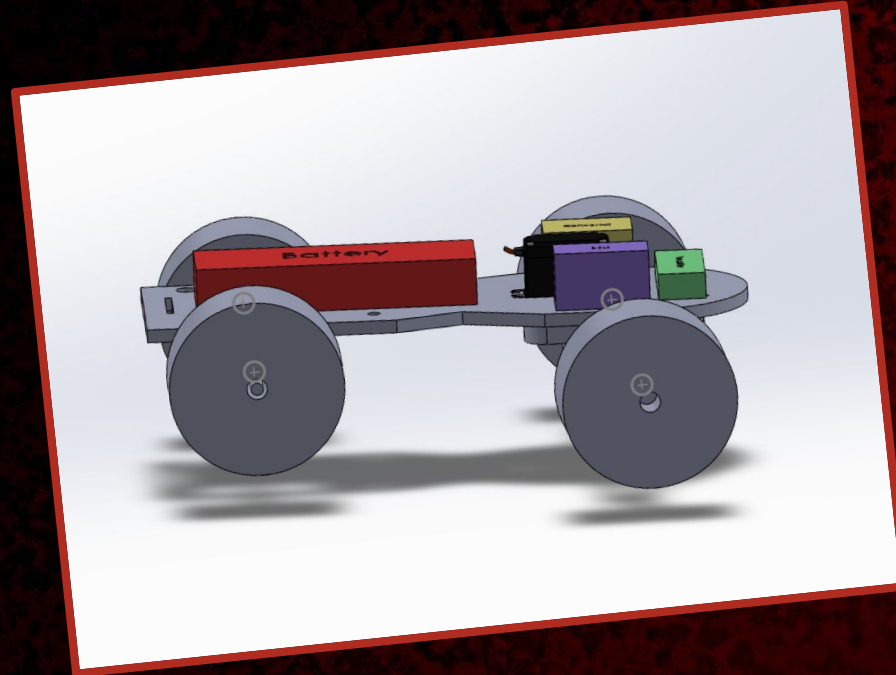
WEEK 1-7 Gantt Chart

[illegible]

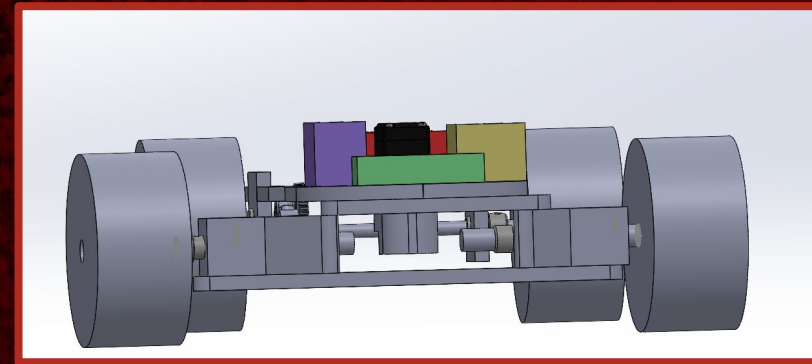
Rover Overview (CAD)



Top View



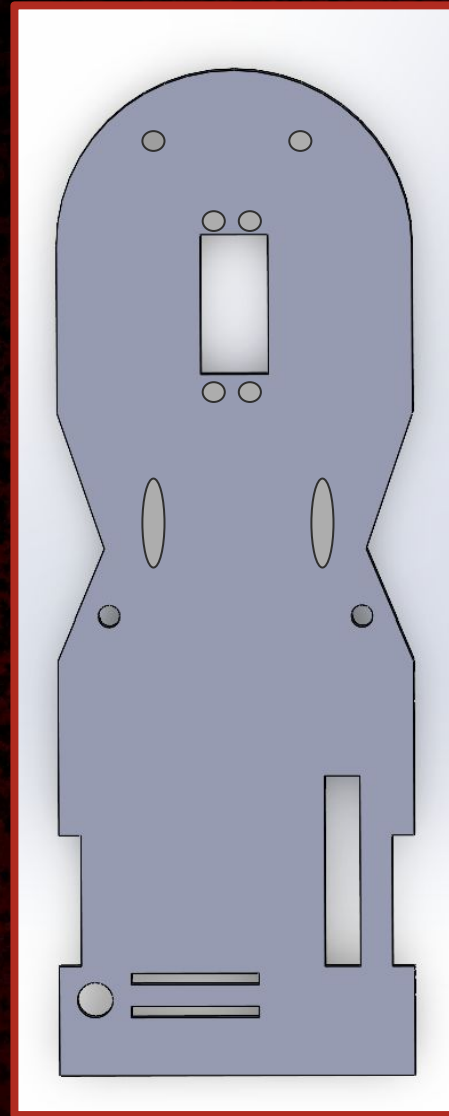
Side View



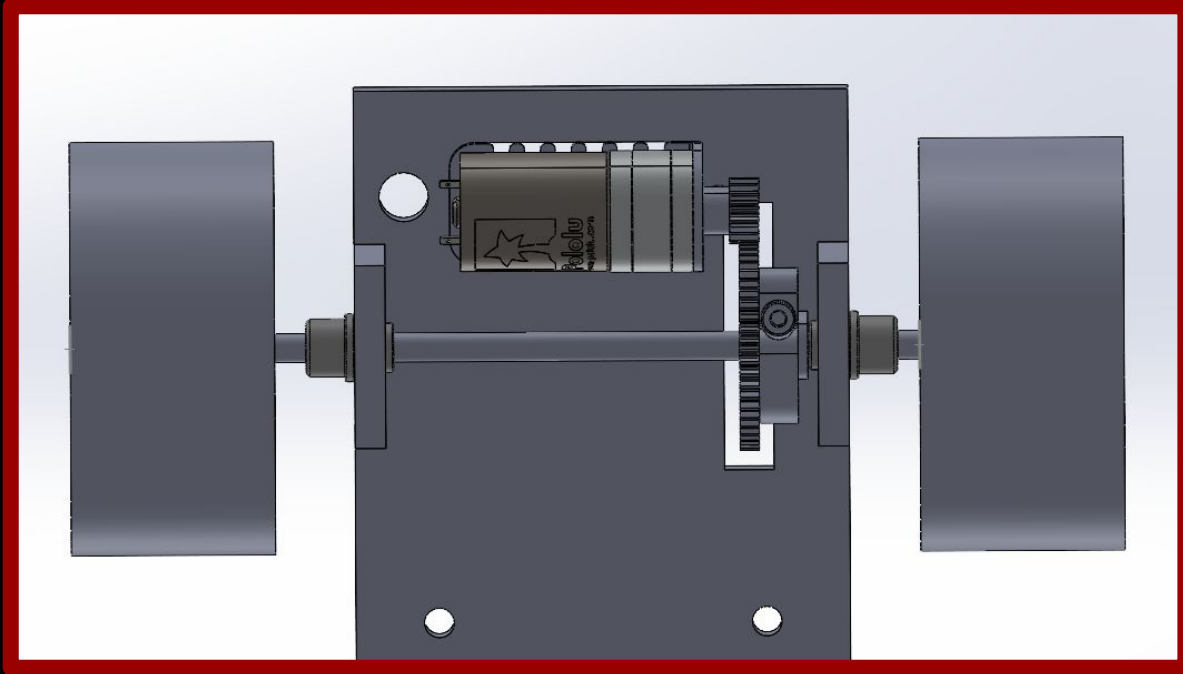
Front View

Chassis Design

- 4" x 11.5" (Width x Length)
- Rounded chassis
 - aerodynamic/less air resistance
- Weight Save



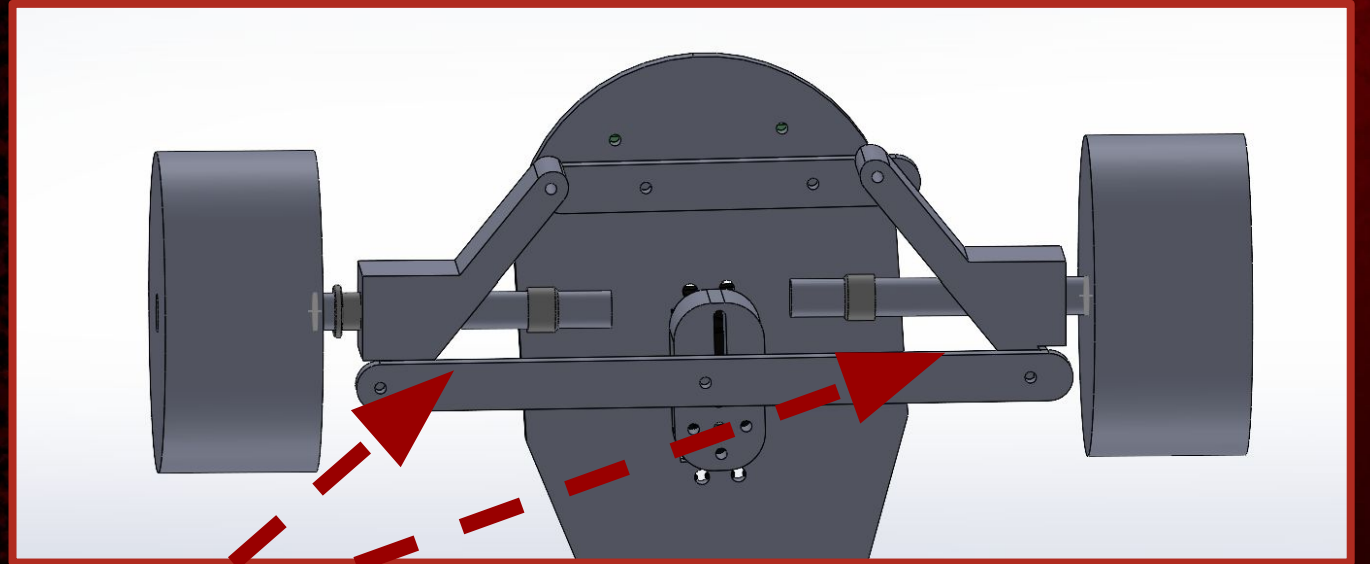
Design Decisions



- 52T Gear
 - Balance between speed and torque
- 85mm Larger Wheels
 - More traction
 - Raised chassis

Steering Design

- Reverse Ackermann Steering
 - Smooth turns
 - Efficiency
 - Increased Stability
- Side linkages at 45° angles



Timeline & Major Milestones (WEEK 8-10)

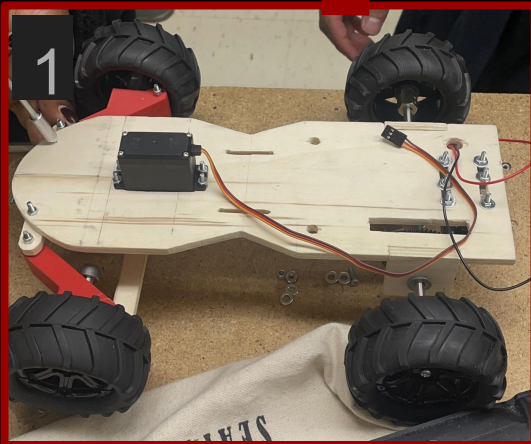
(Week 8)

- Structure Deadline
 - Wheel system
 - gear guard
 - construction
- Begin Electronics

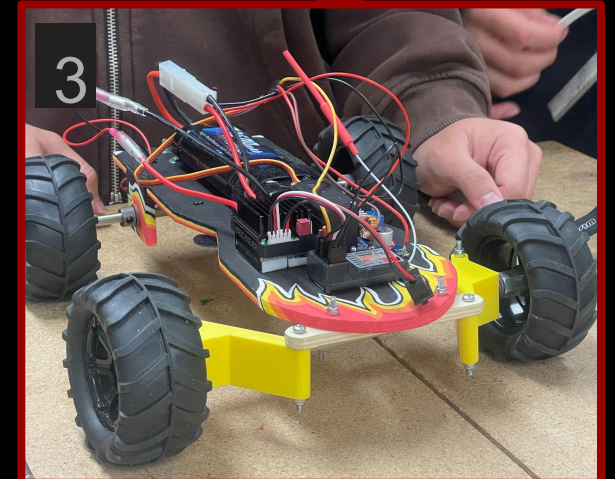


(Week 9)

- Finished electronics
- Finalize Rover



- Order of Fabrication
 - Electronics Malfunctioning
- ## Challenges:



Continued



(Week 10)

- Finished Rover
- Test, Record, Calculate
- Final Presentation

- Several Steering Part Reprints
- Chassis/Parts Adjustments



Lack of Testing time

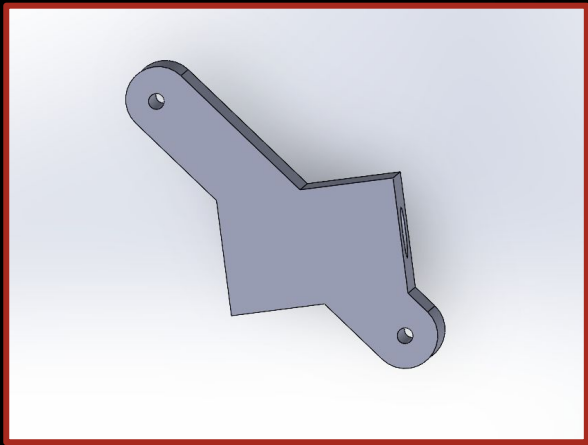


WEEK (8-10) Gantt Chart

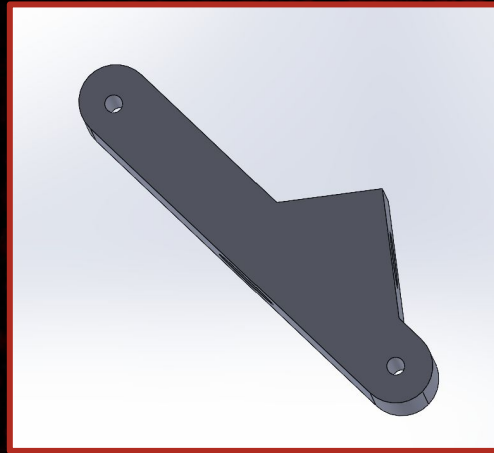
A		AO	AP	AQ	AR	AS	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC	BD	BE	BF	BG	BH
1	Seven Deadly Sins																				
2	Wednesday 3:00pm	Week 8					Week 9					Week 10					Finals Week				
3	Activity	M	Tu	W	Th	F	M	Tu	W	Th	F	M	Tu	W	Th	F	M	Tu	W	Th	F
6	Rover Design																				
7	Determine Rover Layout																				
8	Select Gear Ratio																				
9	Purchase Order Form																				
10	SolidWorks Part Designs																				
11	SolidWorks Assembly																				
12	SolidWorks Detailed Drawings																				
13	Structure Fabrication																				
14	Wheels System																				
15	Motor Mount																				
16	Gear Guard																				
17	Battery holder																				
18	Steering System																				
19	Structure Assembly																				
20	Electrical System																				
21	Mount & Connect Electronics																				
22	Remote Control Tuning for Motor																				
23	Remote Control Tuning for Servo																				
24	Steering System Testing																				
25	Test and Evaluation																				
26	FINAL COMPETITION																				
27	Action Item Reports																				
28	Preliminary Presentation																				
29	Final Presentation																				

Changes during Fabrication:

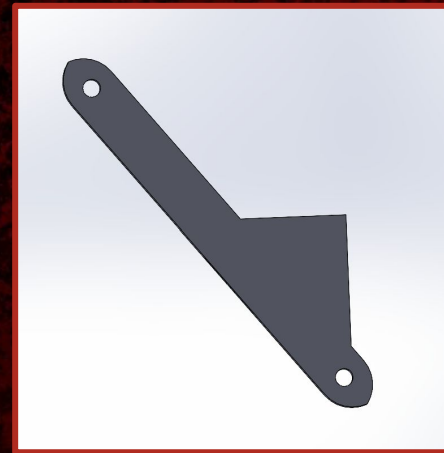
- Side Steering Linkages
 - Re-printed 4 times



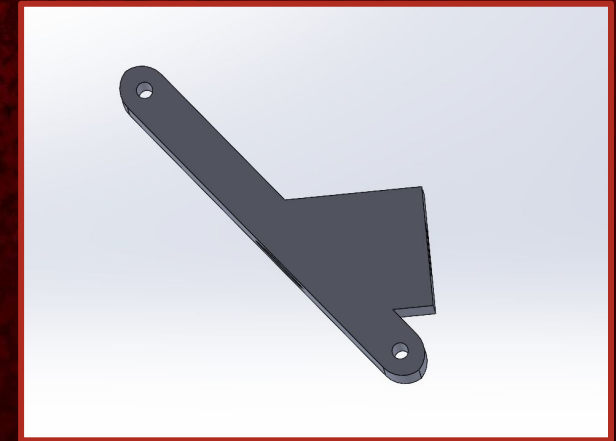
Initial Design



2nd iteration
(Structure
Deadline)



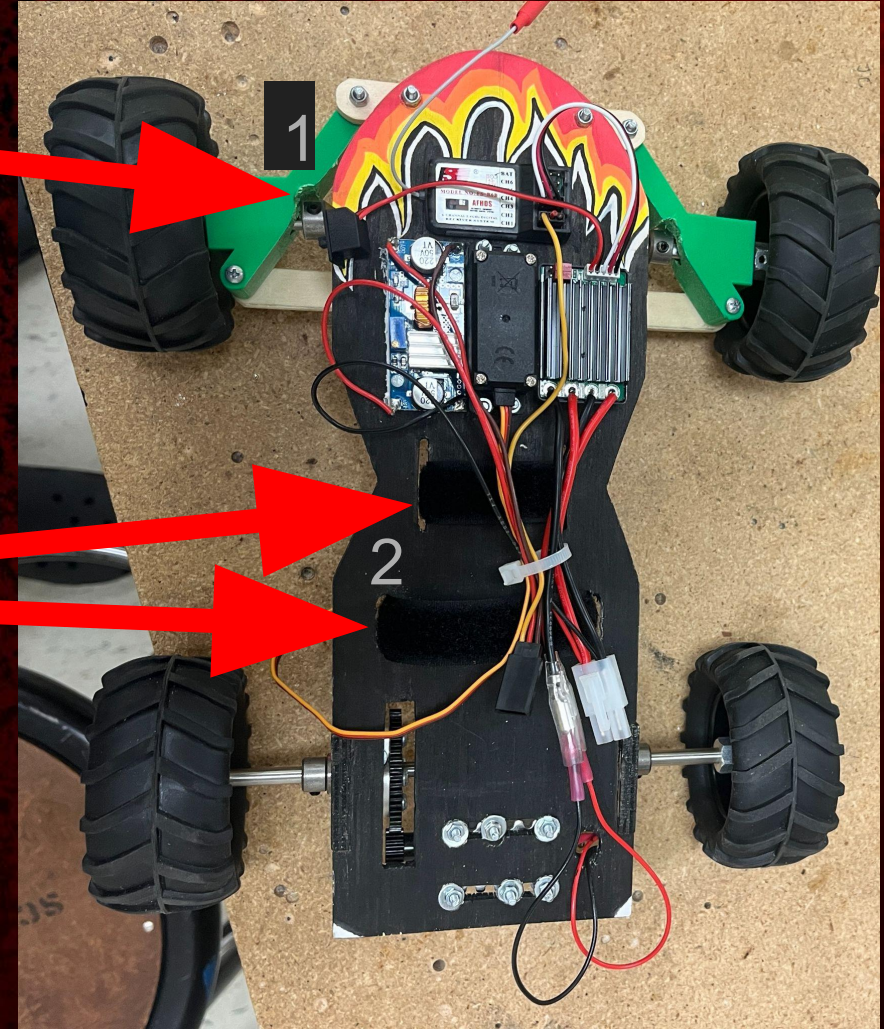
3rd iteration



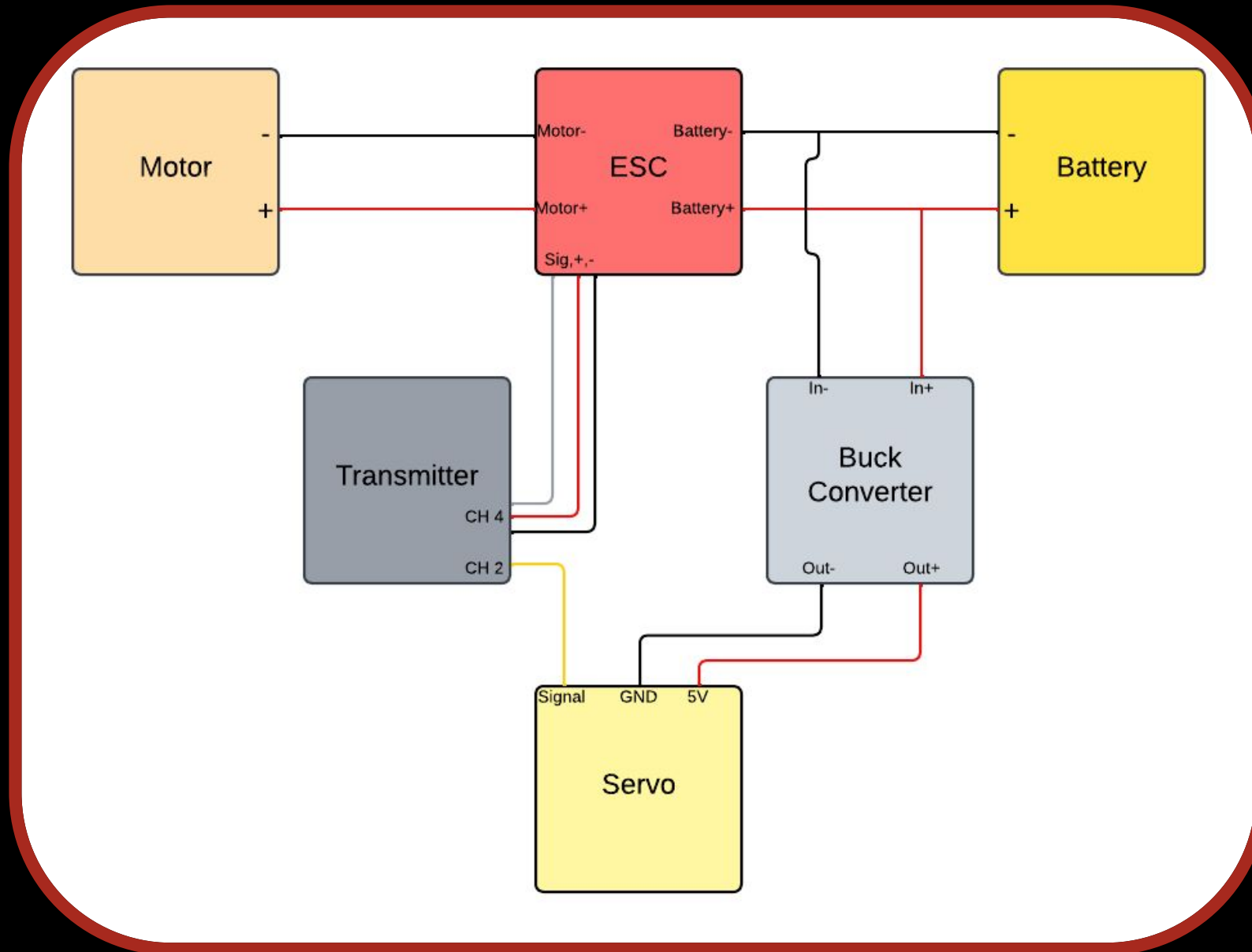
FINAL DESIGN

Changes During Fabrications 2

- Sanded down new printed part to fit rod
- Adjusted holes on chassis to lock in the battery

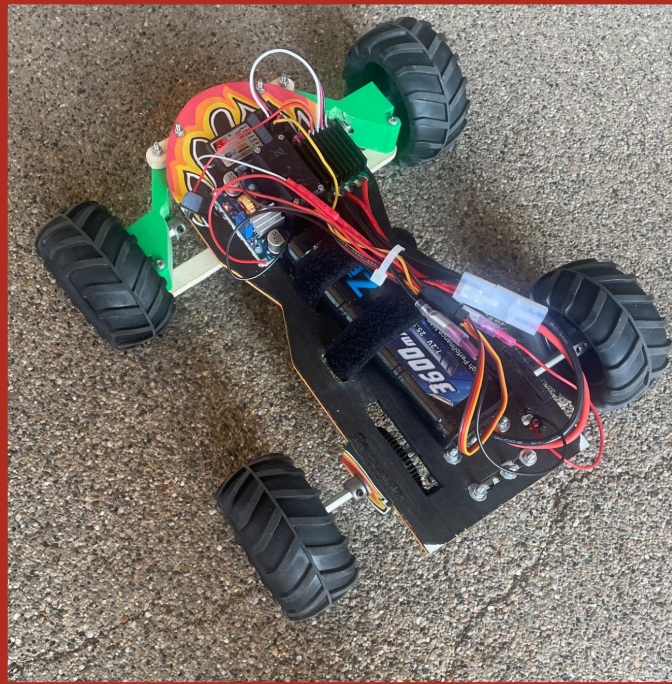


Electronics & Wiring Diagram

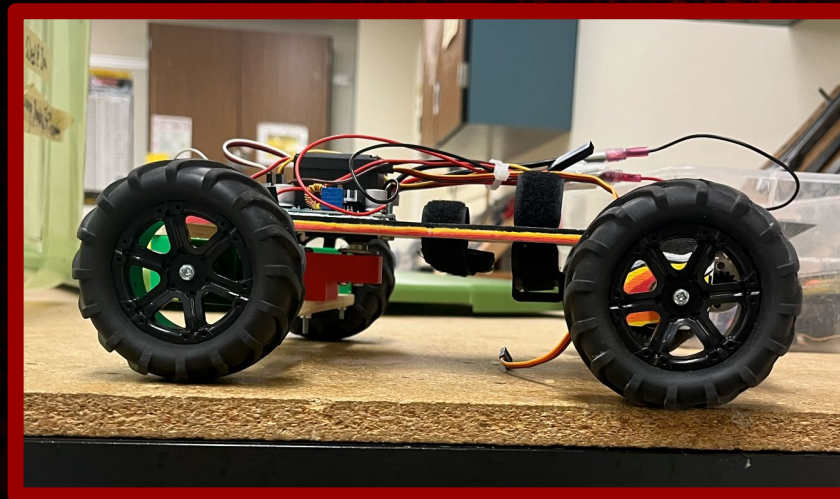


Rover Overview (Final Product)

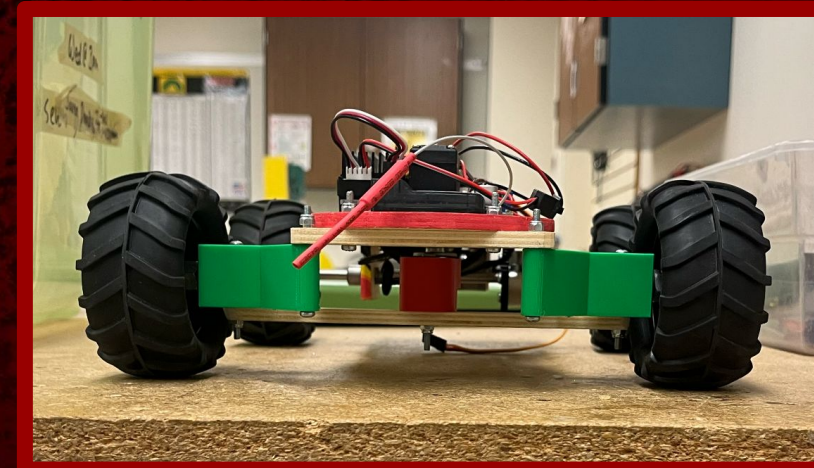
**Weight/Mass of Rover:
1.0721 kg / 2.362 lb**



Top View



Side View

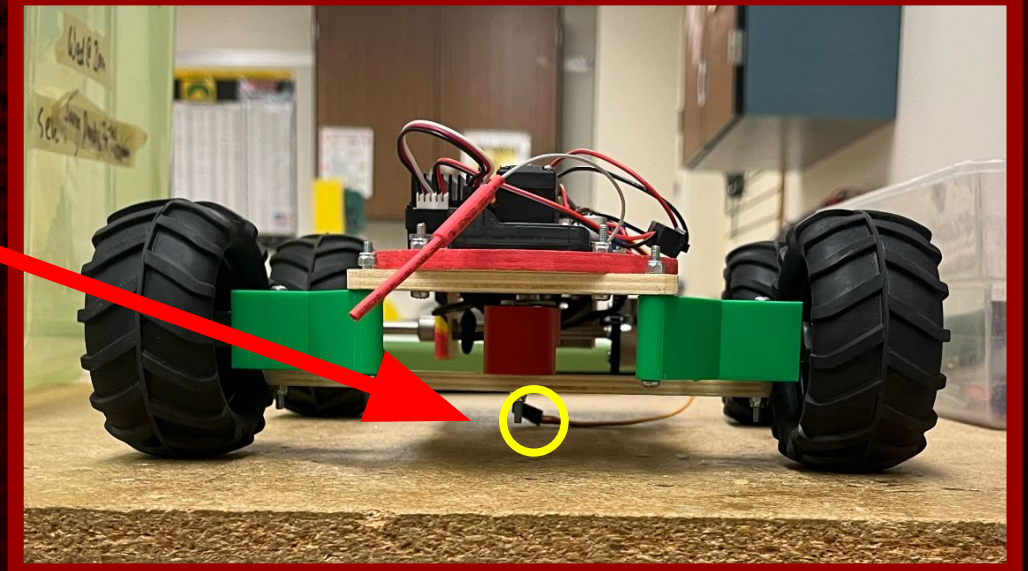
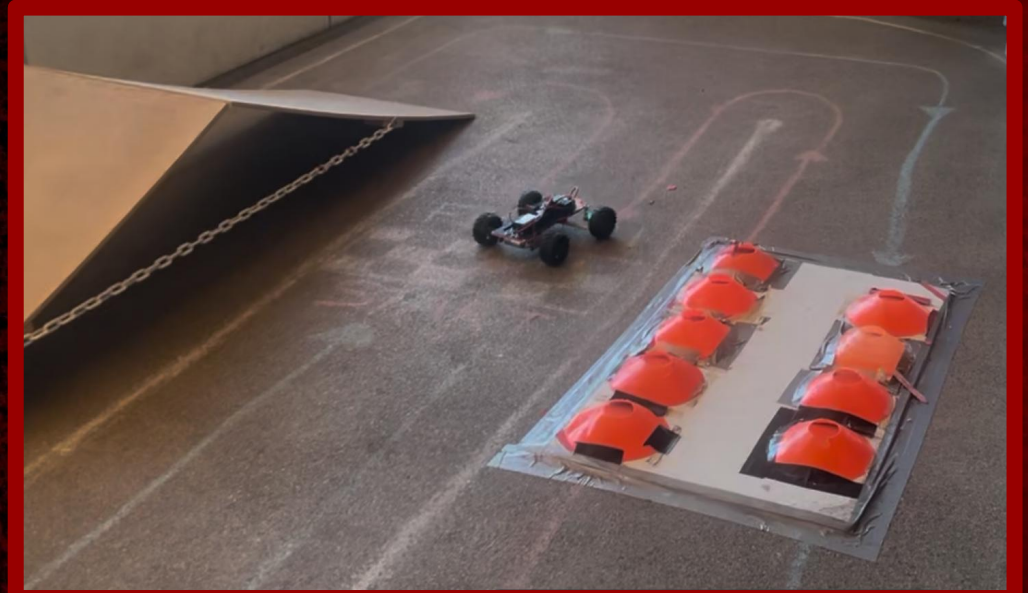


Front View

Rover In Action:

Completed Track: 2m 33s

- Went up inclined ramp
- Struggles on cones due to low clearance
- Struggled due to turning radius



Lessons & Skills Learned

- Teamwork & Leadership Skills
- Hands-on Experience
- Design Skills (CAD)

Testing is important

Don't give up!

DO NOT
replace files on
CAD

Always account for imperfections

End



Thank you!

Questions?